

# NCSU MATH 242 011 - PRACTICE FINAL - FALL 2022

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**Problem 1** (10 points) Let  $\vec{F} = \langle (y + 2z + 3) \sin x, -\cos x, -2 \cos x \rangle$ .

1. Determine if  $\vec{F}$  is conservative. If it is, find all potential functions  $f$  for  $\vec{F}$  so that  $\vec{F} = \nabla f$ .
2. Find the circulation  $\oint_C \vec{F} \cdot d\vec{r}$  where  $C$  is the unit circle.

**Problem 2** (5 points) Parameterize the portion of the paraboloid  $z = 9 - x^2 - y^2$  that lies above  $z = 5$ .

**Problem 3** (10 points) Compute the flux of the vector field  $\vec{F} = \langle x, y, z^2 \rangle$  through the parameterized surface  $S$  given by  $\vec{r}(u, v) = \langle u \cos v, u \sin v, v \rangle$ ,  $0 \leq u \leq 2$  and  $0 \leq v \leq \pi$  with the upward orientation.

**Problem 4** (5 points) Let  $\vec{F} = \langle xye^z, y^2ze^z, xe^z \rangle$ . Compute the divergence  $\nabla \cdot \vec{F}$  and the curl  $\nabla \times \vec{F}$ .

**Problem 5** (10 points) Compute the surface integral of the function  $f(x, y, z) = (x^2 + y^2)z$  over the portion of the sphere  $x^2 + y^2 + z^2 = 9$  that lies in the first octant.

**Problem 6** (10 points) Using Greene's theorem and the vector field  $\vec{F} = \langle -y, x \rangle$ , find the area bounded by the parametric curve  $\vec{r}(t) = \langle \sin(2t), 2\sin(t) \rangle$  with  $0 \leq t \leq \pi$ .

**Problem 7** (10 points) Verify Stokes Theorem for  $\vec{F} = \langle -y, x, 2 \rangle$  with  $\nabla \times \vec{F} = \langle 0, 0, 2 \rangle$ , where  $S$  is the cone  $x^2 + y^2 = z^2$  with  $0 \leq z \leq 4$ .

**Problem 8** (5 points) Use the Divergence Theorem to set up a triple integral that calculates the flux integral  $\iint_S \vec{F} \cdot \hat{n} dS$  where  $\hat{n}$  is outward pointing, for  $\vec{F} = \langle x^2 + \sin(yz), y - xe^{-z}, z^2 \rangle$  and  $S$  is the surface of the solid bounded by the cylinder  $x^2 + y^2 = 4$  and the planes  $z = 0$  and  $x + z = 2$ . Do not evaluate.

**Problem 9** (10 points)

1. Find the tangent plane of  $f(x, y, z) = e^{x^2+y^2+z^2}$  at the point  $(1, 1, 1)$
2. Write the linearization (aka linear approximation) of  $f(x, y, z)$  near the point  $(1, 1, 1)$

**Problem 10** (10 points) Find the global extreme values of  $f(x, y) = x^2 - xy + y^2 + 1$  on the closed triangular region in the first quadrant bounded by the lines  $x = 4$ ,  $y = 0$ , and  $y = x$ .

**Problem 11** (10 points) Use Lagrange multiplier techniques to find the local extreme values of  $f(x, y) = x^2 + y^2$  subject to the constraint  $2x^2 + 3xy + 2y^2 = 7$ .

**Problem 12** (5 points) Compute the directional derivative in the direction of the function  $f$  given by  $f(x, y) = x^2y^4$  at  $(2, 1)$  in the direction from  $(0, 0)$  to  $(5, 1)$ .